fuel)
Cslip
/Cfug
(mas
s %)

ec
gCO
2eq/g
fuel
TtW
GHG
intensity
(gCO2eq
/MJ)
NOTE
105 Hydrogen H2_f_SMR_CCS_gm 0.12 ALL ICEs 0 Fuel cell
121 Ammonia NH3_rN2_fH2_HB_gm 0.0186 ALL ICEs 0 Fuel cell

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SOURCE: MEPC.376(80).pdf
121 Ammonia NH3_rN2_fH2_HB_gm 0.0186 ALL ICEs 0 Fuel cell

RESOLUTION MEPC.376(80) (adopted on 7 July 2023)
GUIDELINES ON LIFE CYCLE GHG INTENSITY OF MARINE FUELS (LCA GUIDELINES) MEPC 80/17/Add.1
Annex 14, page 54

APPENDIX 3

ABBREVIATIONS AND GLOSSARY

Abbreviations

AR IPCC Assessment Report
BDN Bunkering Delivery Note
Cf Emission conversion factors C fCO2/CH4/N2O (g GHG (CO 2/CH4/N2O)/g fuel) for emissions of the combustion and/or oxidation process, including the fuel with relevant GWP effect resulting from the combustion energy conversion
CH4 Methane
CO2 Carbon dioxide
CO2eq Carbon dioxide equivalent
CCS Carbon Capture and Storage
CCU Carbon Capture and Utilization
DAC Direct Air Capture
DCS IMO ship fuel oil consumption Data Collection System

DLUC Direct Land Use Change
FLL Fuel Lifecycle Label
GHG Greenhouse gas
GWP Global Warming Potential
ILUC Indirect Land Use Change
IPCC Intergovernmental Panel on Climate Change
LCA Life Cycle Assessment
LCV Lower Calorific Value (MJ/g fuel)
NMVOC Non-Methane Volatile Organic Compounds
N2O Nitrous oxide

---CHUNK---

SOURCE: MEPC.376(80).pdf
ILUC Indirect Land Use Change
IPCC Intergovernmental Panel on Climate Change
LCA Life Cycle Assessment
LCV Lower Calorific Value (MJ/g fuel)
NMVOC Non-Methane Volatile Organic Compounds
N2O Nitrous oxide
NTC NOx Technical Code
RFNBO Renewable Fuels of Non-Biological Origin
SLCF Short-Lived Climate Forcers
TtW Tank-to-Wake
WtT Well-to-Tank
WtW Well-to-Wake
VOC Volatile Organic Compounds

Glossary

Co-product an outcome of a production process, which has a relevant economic value and elastic supply (intended as the existence of a clear evidence of the causal link between feedstock market value and the quantity of feedstock that can be produced).

Biomass Biomass is renewable organic material that comes from plants and animals.

Renewables any form of energy from solar, geophysical or biological sources that is replenished by natural processes at a rate that equals or exceeds its rate of use. Renewables are obtained from the continuing or repetitive flows of energy occurring in the natural environment and includes low-carbon technologies such as solar energy, hydropower, wind, tide and waves and ocean

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SOURCE: MEPC.376(80).pdf
from the continuing or repetitive flows of energy occurring in the natural environment and includes low-carbon technologies such as solar energy, hydropower, wind, tide and waves and ocean thermal energy, as well as renewable fuels such as biomass.

Global Warming Potential Global Warming Potential indicates the potential of a greenhouse gas to trap extra heat in the atmosphere over time in relation to carbon dioxide. The enhanced heat trapping in the atmosphere (i.e. the "greenhouse effect") is caused by the absorption of infrared radiation by a given gas. The GWP also depends on the atmospheric lifetime of a gas, and the

time-horizon considered (for example, GWP20 is based on the energy absorbed over 20 years, whereas GWP100 is based on the energy absorbed over 100 years. Each greenhouse gas has a specific global warming potential which is used to calculate the CO2-equivalent (CO2eq).

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SOURCE: MEPC.376(80).pdf
whereas GWP100 is based on the energy absorbed over 100 years. Each greenhouse gas has a specific global warming potential which is used to calculate the CO2-equivalent (CO2eq).

Land Use Change - Production of bio-based fuels leads to land use change (LUC). LUC can be classified as direct LUC (DLUC) and indirect LUC (ILUC).

Life Cycle Assessment (LCA) framework Life Cycle Assessment determines the potential environmental impacts of products, processes or services from cradle to grave, e.g. from acquisition/extraction of raw materials through to processing, transport, use and disposal.

System boundaries The system boundary determines which entities (unit processes) are inside the system and which are outside. It essentially determines which life cycle/supply chain stages and processes are included in the assessment and need to be in accordance with the goal and scope of the study.

System expansion ISO 14040 recommends the use of system expansion whenever possible. System expansion is part of the consequential LCA method that seeks to capture change in environmental impact as a consequence of a certain activity.

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SOURCE: MEPC.376(80).pdf
System expansion ISO 14040 recommends the use of system expansion whenever possible. System expansion is part of the consequential LCA method that seeks to capture change in environmental impact as a consequence of a certain activity.

Well-to-Wake - WtW studies estimate the energy requirements and the resulting greenhouse gas (GHG) emissions in the production of a fuel and its use in a ship, based on the broader Life Cycle Assessment (LCA) methodology. The term 'Well' is used for fuels from all sou rces, because although the term is most applicable to conventional crude oil resources, it is widely used and understood.

APPENDIX 4

TEMPLATE FOR WELL-TO-TANK DEFAULT EMISSION FACTOR SUBMISSION

1 Explanatory remarks on the general scope of the template: This template aims at collecting and presenting in a clear and structured manner the input data used to calculate a "default

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SOURCE: MEPC.376(80).pdf
1 Explanatory remarks on the general scope of the template: This template aims at collecting and presenting in a clear and structured manner the input data used to calculate a "default emission factor" for a specific "feedstock -to-fuel" pathway. A "default emission factor" aims as representing the quantitative results of a high-level assessment about the carbon intensity (gCO2eq/MJ) of a feedstock-to-fuel value chain. The default emission factor is not meant to represent the best available way to produce a fuel, rather a value potentially describing a feedstock production then converted in a standard plant, located in a generic region. A default emission factor does not have to capture process improvement, with respect to current production, nor innovative technologies. The goal of default emission factor is, at least, twofold:

.1 allow for a carbon intensity comparison among different technologies;

.2 allow for operators to demonstrate lower core life cycle emissions compared to the default core life cycle, thought a certification process.

2 Operators (e.g. fuel producers) can ask for being certified, in order to prove better